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Line Following Robot System

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We would like to convey our sincerest thanks to our module instructor, Mr. Jagannath Paudyal, who has provided us with valuable advice, continuous support, and helpful suggestions for the project. He is an expert in Cloud Computing and IoT and has been able to help us realize the real-life applications of embedded systems in designing the line following robot.

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Abstract

This study report on design and construction of an autonomous line-following robot using components such as the Arduino Uno board, two IR sensors, a motor driver, and two DC motors. It has been developed to follow a Black line on a white surface independently and is intended to reduce inefficiencies in the manual handling of materials in various industries.

It has been made possible using a very basic control mechanism whereby the line is sensed by the IR sensors, analysed by the Arduino board, and then used by the motor driver to drive the movement of the robot to go forward, left, right, or stop. The hardware was assembled on a chassis and powered by a battery. Simulations were carried out on circuit designer, and real-life experiments were performed on the straight path, curves, and T-junction.

Five tests conducted were linear motion, offset correction, making sharp turn, T junction halt, and IoT sensor output simulation. The robot did well in tracking the line at moderate speeds. Sharp turns were done well when the speed was reduced. The IoT simulation test proves that sensor data can be transmitted in readiness for cloud-based processing.

The project achieves the goal of making decisions based on real-time analysis, integration of sensors, and implementation of IoT basics for low-cost automation.

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1. Introduction

A line following car is a traditional path tracking robot, which drives along the settled path without any hand control, typically marked by a black line on a white surface (Luo, 2023). This type of robot uses sensors that detect the position of the line and send signals to the microcontroller, where the data is processed which controls the movement of the motors in the robot. This allows the vehicle to move without any human intervention. This robot has been evolving since the 1950's from a basic path following to complex control systems. Such systems are widely applied in various fields, including industrial automation, logistics distribution, home services and healthcare (Shi, 2026).

This project focuses on the development of a line following car that utilizes key concepts from electronics, robotics and embedded systems using both hardware and software that can work together to create an intelligent system. This project demonstrates real time decision-making, automation and control system in a simple and effective way along with highlighting the importance of precision, adaptability and efficiency in an automated system.

1.1 Current Scenario

In this modern age and era robotics and automation have been rapidly advancing becoming an essential in industries like manufacturing, logistics and transportation. Systems like line following car has been widely used to perform repetitive and time consuming work with greater efficiency and accuracy than humans. Line following robots are used widely in automated guided vehicles in warehouse and factories to transport and move things around. Line following robots can lift heavy objects and transport them without fatiguing humans. These types of system helps the company to reduce labour cost, minimize human error and improve productivity.



Figure 1: Line following robot working in a industry

1.2 Problem Statement

In this modern era and environment, the manual movement of goods and services while navigating along the fixed routes can be inefficient, time-consuming and can lead to human fatigue and error especially in industrial setting. Moving goods manually by a human can cause many challenges such as fatigue, inconsistency and reduced productivity while also being very time consuming. Manual system also lacks precision and can lead delay or accidents in the process. In industrial setting where goods are constantly being moved, an automated solution can be reliable. A line following robot can follow the designated path with high accuracy and minimal supervision which can increase the company's overall productivity. This project aims to address these issue faced by many industries by developing a line following car that can follow the designated path with accuracy.

1.3 Aims and objectives

Aims:

The aim of this project is the designing and development of a line follower car which can travel on a specified path based on input from sensors, microcontroller and IoT implementation.

Objectives:

- To design and develop the system which involves sensors, microcontroller and motor driver circuitry.
- To incorporate and develop IR sensor to detect a pre-defined path.
- To design control logic for moving car on the specified path in a smooth way.
- To integrate and assemble all hardware parts properly.
- To enhance the performance of the developed system.
- To consider basic aspects of IoT into the project.

2. Background and System Design

2.1 Required Analysis

The line following car operates using the response of the IR sensor and with real-time decision making through Arduino Uno board. The system processes the signals of IR sensor in real time to influence the direction of the motor.

The control logic is affected by the sensors readings which influences the movement of the vehicle, and the response time depends on the speed of sensor detection and processing efficiency of the microcontroller. The performance of the system is affected by the factors such as lighting, surface conditions and sensor calibration. These external variables can affect the robot's direction and speed which can also lead to inaccurate readings, causing deviation from the path. Another limitation is the lack of adaptive decision-making, the robot only follows a fixed logic and cannot optimize its path.

2.2 Individual Contribution

Student Name	Primary Role	Specific Responsibilities
Khushi Neupane	Project Lead & Hardware Integration	Overall coordination, chassis assembly, motor and wheel mounting, power distribution, writing most of the Arduino code
Prasansha Yadav	Documentation & Core Testing	Report writing, test track design, executing test cases, documenting results, assisting with code debugging
Priya Thapa	Simulation & IoT Integration	Circuit designer for circuit simulation, system diagrams (Draw.io), researching basic IoT, formatting the final document
Dipsha Gurung	Sensor Setup	Assisted in placing IR sensors on the chassis and adjusting their height; attended meetings; reviewed the final report for spelling errors
Lishara Rana	Battery Connection	Helped connect the battery clip to the motor driver; checked the reference format, attended meetings, contributed in planning.

Table 1: Individual Contribution

2.3 System Overview

The line following car system is designed with sensor, microcontroller and actuator that works together to achieve autonomous navigation along the settled path. This line following car consists of Infrared (IR) sensors as the sensor, Arduino uno as the microcontroller and DC Motor as an actuator along with a motor driver and batteries as a power supply. Each of these components play a role in ensuring that the project is accurate and efficient.

The working of the system begins with two IR sensor which can be used to detect light, each of these IR sensors are placed on the left and the right sides of the robot to detect the line respectively. These IR sensors then differentiate between the black line and the white background based on the light reflection which generates digital or analog signal. Then the sensor data and input is transmitted to the microcontroller which processes in real time which determines the movement of the car (Muhammad Ahmad Baballe, 2023).

After the data is processed the microcontroller sends the signals to motor driver which is acts as an interface between power control signals by controlling the speed and direction of the car. The entire system is powered by a battery.

2.4 System Design and Architectural Diagrams

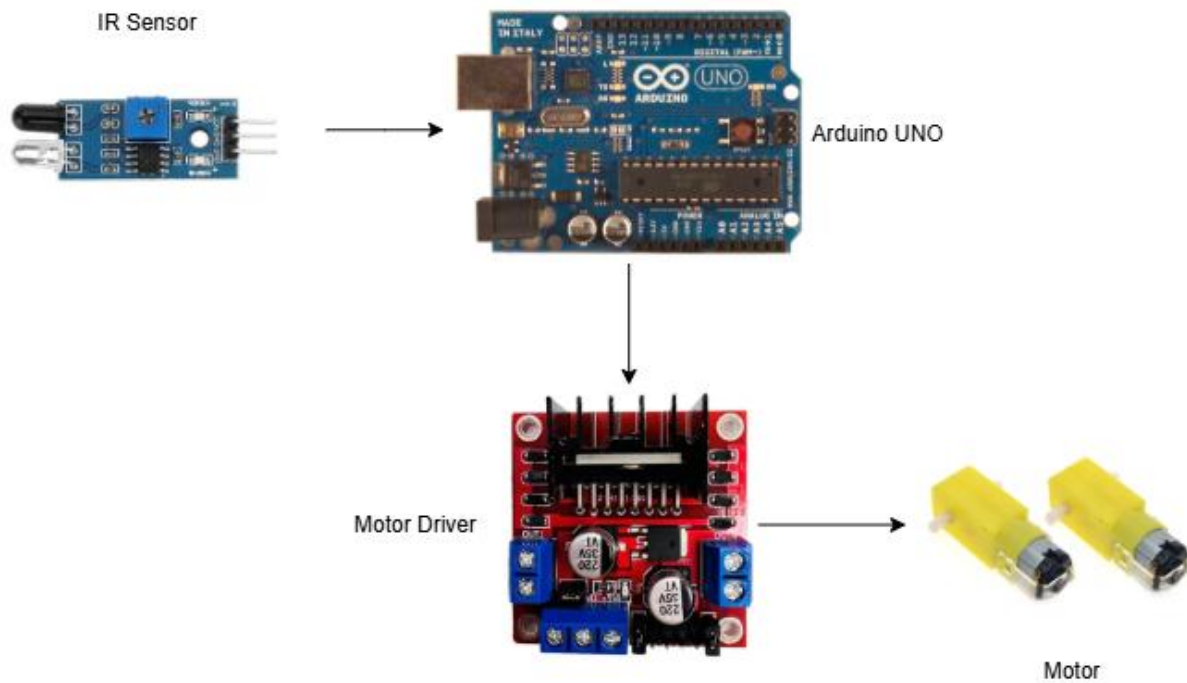


Figure 2 System and Architectural Diagrams of line following car

2.5 Block Diagram

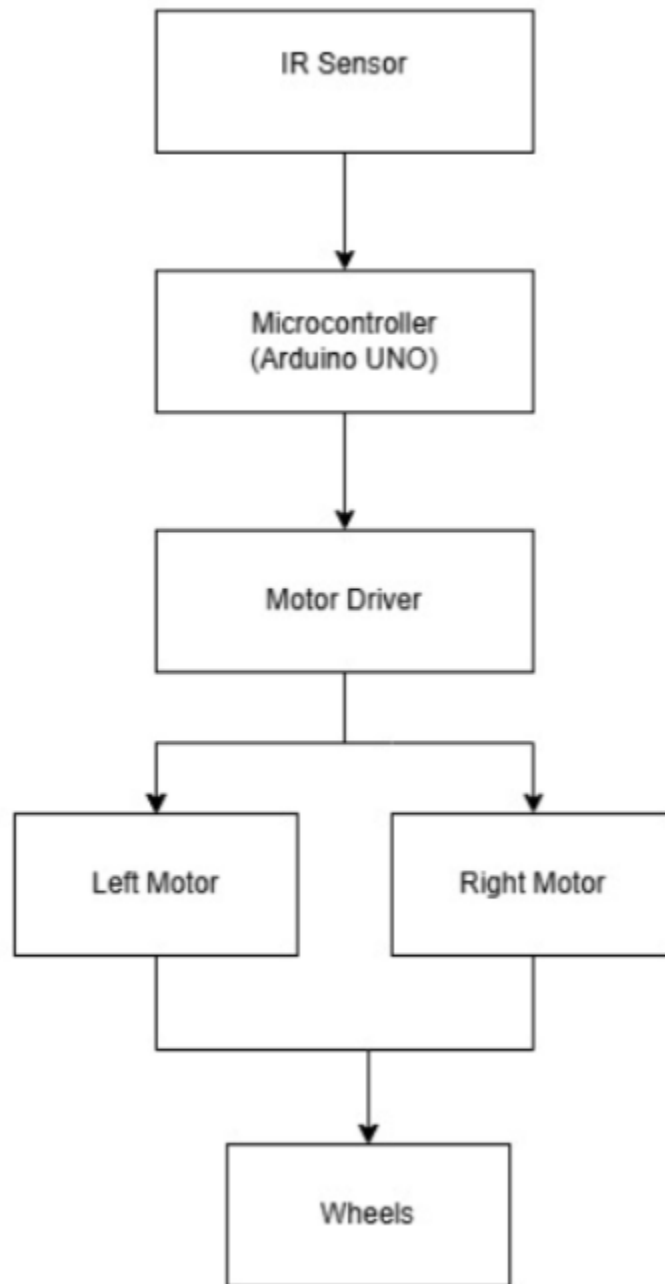


Figure 3: Block diagram of a line following car

2.6 Circuit Diagram

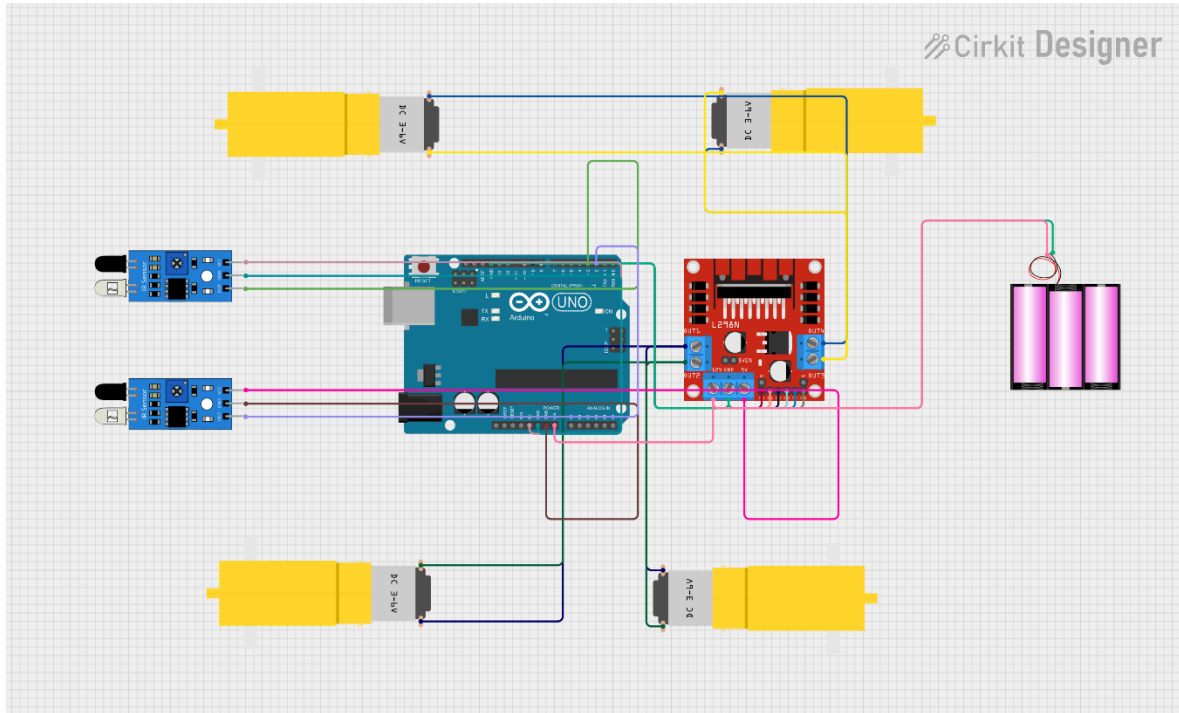


Figure 4 Circuit Diagram



2.8 Detailed Analysis

2.8.1 Hardware Components

1. Arduino Uno (Microcontroller)

Arduino UNO is an open-source microcontroller which has an Atmega328P microprocessor, and this forms the brain of the Arduino UNO microcontroller circuit.

This is the brain of our car. It gets input from the sensors, processes it and gives instructions to the motors. In essence, it determines what direction the car should move (Hemmings, 2018).



Figure 6: Arduino Uno Board (Microcontrollers, 2026)

2. IR Sensors (Infrared Sensors)

IR Sensor is an electronic component which uses Infrared sensing technology to detect any object in its surroundings. IR Sensors detect various characteristics like the movement of an object or temperature change (Efyian, 2026).

Here,

- IR sensors are used for detecting the black line.
- These send out the Infrared signals.
- In case there is a white surface, then the light rays are reflected so sensor outputs HIGH signal.
- In case of black line, the light rays are absorbed so the sensor outputs LOW signal.



Figure 7: IR Sensor (Harish Projects, 2026)

3. Motor Driver Module

Motor driver also referred to as the control motor. The function of the motor driver is to control and manage the operations of the electric motor. This electronic equipment forms the link between the microcontroller or any other control system and the electric motor to ensure proper motor performance (PicoBricks, 2023).

Here, the Arduino cannot directly power the motors. The motor driver acts as a central controller:

- It receives commands from the Arduino board
- It manages motor rotation (clockwise/ counter-clockwise)
- It supplies sufficient current to drive the motors

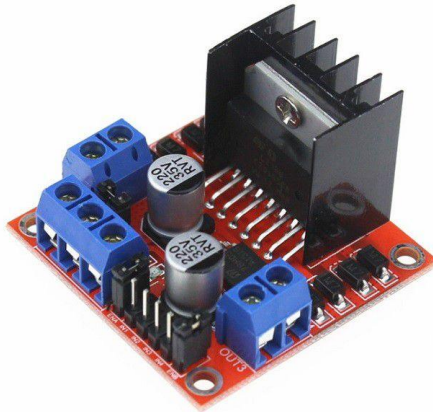


Figure 8: Motor Driver (Cytron, 2026)

4. DC Motor

DC Motor is an electric device that turns electrical energy to mechanical energy used by DC motor is DC and it turns into mechanical rotation.

Here,

- They are what really make the wheels rotate.
- Motor drivers control speed and direction of rotation according to the instructions from Arduino.



Figure 9: DC Motor (Magnetic Innovations, 2026)

5. Chassis and Wheels

The car chassis is also called the frame, but the chassis is only the portion that carries the weight of the care frame. The chassis gives the vehicle the stability required to make the car strong (formula one autocentres, 2023).

Here,

- This is your car body.
- It contains all the parts and gives you mobility. Here stability is essential for good line following.



Figure 10: Chassis and Wheels (Wiltronics, 2026)

6. Power Supply (Battery)

Power sources are an essential part of the Internet of Things (IoT) system. The devices used generate the necessary electrical energy required to power the different parts (Emergy Tech, 2024).

Here,

- It powers the whole system i.e. Arduino and motors.
- The car could become uncontrollable without a stable power source.



Figure 11: Battery (Daraz, 2026)

7. Jumper Wires

Jumper wires are wires with connectors on both ends which enables you to join two places using the jumper wires without any need for soldering them. Jumpers are usually used when working with the breadboard or other kinds of prototyping equipment to make it easier to modify your circuit (Hemmings, 2018).

Here, it's used for connecting all the circuit elements to one another.

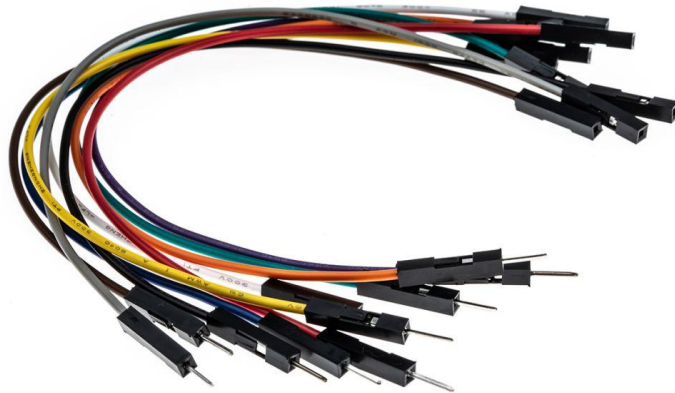


Figure 12: Jumper Wires (RS, 2026)

2.8.2 Software Components

1. Arduino IDE

Programming the Arduino requires a software in which the code will be written and uploaded. This is done using the Arduino IDE (GeeksforGeeks, 2025).

Here, this software is used to write and upload the code to the Arduino.



Figure 13: Arduino IDE (Tecnock, 2026)

2. Draw.io

Draw.io, which is also known as diagrams.net, is an online drawing tool that is used to draw flowcharts, network diagrams, UML diagrams, organizational charts, and floor plans (draw.io, 2025).

Here, it has been used to design the system architecture block diagram and flowchart.



Figure 14: Draw.io (G2, 2026)

3. Simulation Tool (CirKit Designer)

CirKit Designer is an integrated development environment (IDE) that allows users to design electronic circuits using artificial intelligence (AI). The web-based software was created with the purpose of prototyping electronics, and it mainly concentrates on microcontroller boards such as Arduino, ESP32, and Raspberry Pi (CirKit Design , 2026).

In this project, we will use it for:

1. Designing the circuit layout
2. Visualizing component connections
3. Simulating the IoT-based circuit setup
4. Checking wiring and hardware integration

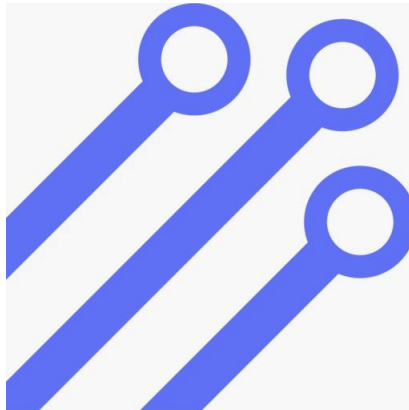


Figure 15: CirKit Designer (CirKit Design , 2026)

4. Simulation Tool (Tinkercad)

Tinkercad is a web application developed by Autodesk that serves as a simple tool to design three-dimensional models and electronics circuits, as well as to learn programming with blocks (Boston Public Library, 2025).

In this project, we will use it for:

1. Simulating the circuit
2. Testing the code
3. Checking connections
4. Debugging



Figure 16: Tinkercad (Merlot, 2026)

3. Development Process

The development process of the project is based on the Agile and prototyping methodology. Thus, the creation of the system will be conducted in iterations, where first, the specification and design of the system is established, and second, the development of the system components takes place.

At each iteration, the prototype will be constructed and improved upon through testing and evaluation of performance. The last step in the process involves integrating all components together into a functional prototype.

Once completed, the prototype will be evaluated to ensure it satisfies the needs of the project.

3.1 Hardware Process:

Step 1: Prepare the Chassis

1. Take the plastic car chassis kit
2. Attach four gear motors on the front and the back of the car using screws and nuts

Step 2: Mount the Wheels

1. Attach wheels on the motor shafts
2. Put the top chassis and tighten it to the set

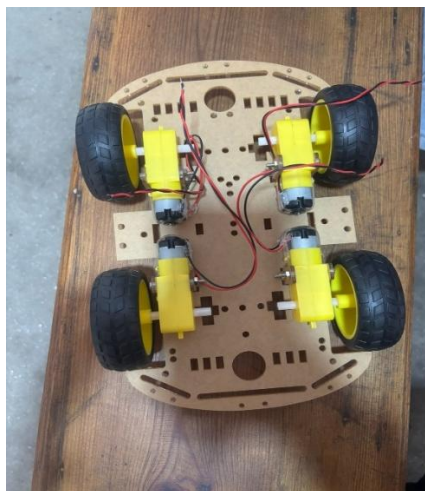


Figure 17: Attaching wheels and gear motors

Step 3: Mount the Arduino Uno board

1. Place the Arduino Uno board on the top of the top chassis
2. Secure it with double sided tape
3. Make sure the USB port of the Arduino board is accessible

Step 4: Mount the Motor Driver

1. Place the Motor Driver module next to the Arduino Uno board
2. Secure it with double sided tape

Step 5: Mount the IR Sensors

1. Take 2 IR sensor modules
2. Attach them to the front of the chassis using double sided tape
3. Space them out about 2-4cm apart
4. Make sure that the sensor legs are facing down in the ground
5. Sensor legs should be 1-2 cm above the ground

Step 6: Connect Wires from gear motors to Motor Driver

1. Left motor wires:
Motor + to OUT1 and Motor – to OUT2
2. Right motor wires:
Motor + to OUT3 and Motor – to OUT4

Step 7: Connect wires from Motor driver to Arduino board

1. Connect ENA pin of Motor Driver to pin 11 of Arduino
2. Connect IN1 pin of Motor Driver to pin 10 of Arduino
3. Connect IN2 pin of Motor Driver to pin 9 of Arduino
4. Connect IN3 pin of Motor Driver to pin 8 of Arduino
5. Connect IN4 pin of Motor Driver to pin 7 of Arduino
6. Connect ENB pin of Motor Driver to pin 6 of Arduino
7. Connect GND pin of Motor Driver to GND pin of Arduino
8. Connect 12v pin of Motor Driver to VIN pin of Arduino

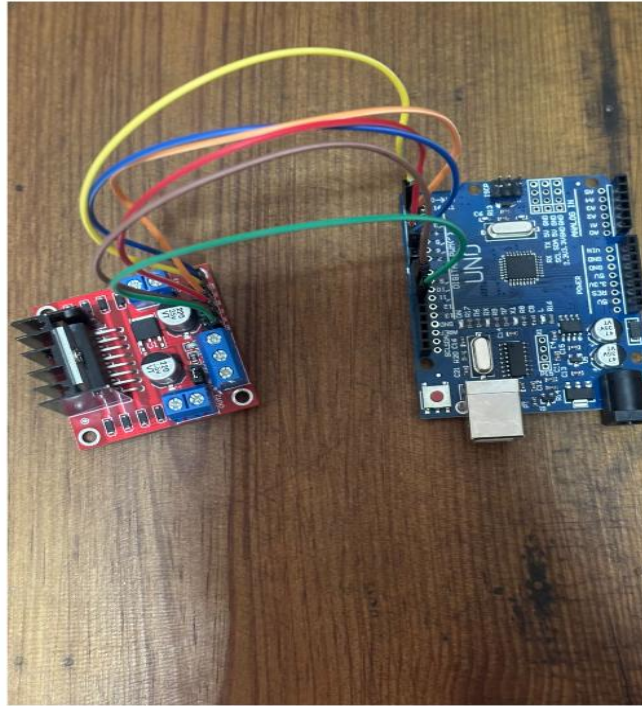


Figure 18: Connecting wires from motor driver to arduino

Step 8: Connect IR sensors to Arduino and Motor driver

1. Connect VCC of right IR sensor to 5v pin in Arduino
2. Connect GND of right IR sensor to GND pin of Arduino
3. Connect OUT of right IR sensor to pin 3 of Arduino
4. Connect VCC of left IR sensor to 5v pin in Motor Driver
5. Connect GND of left IR sensor to GND pin of Arduino
6. Connect OUT of left IR sensor to pin 2 of Arduino

Step 9: Connect Power

1. Use a rechargeable battery to power the vehicle
2. Connect the Battery + to 12v in Motor driver
3. Connect the Battery – to GND in Motor driver

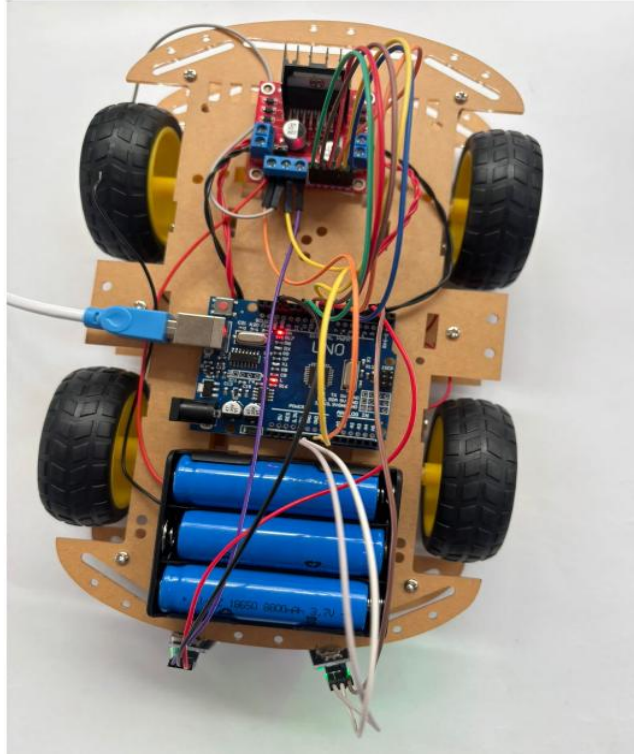


Figure 19: Final Product

Step 10: Upload the Code

1. Open Arduino IDE and display the code
2. Select the board as Arduino Uno and correct COM port
3. Connect Arduino to PC using the USB cable
4. Click Upload

Step 11: Test Sensor

1. Open the Serial Monitor in IDE and check the signal sent by IR sensors
2. Adjust the potentiometer of each sensor to show 0 0 if the surface is white
3. Adjust the potentiometer of each sensor to show 1 1 if the surface is black

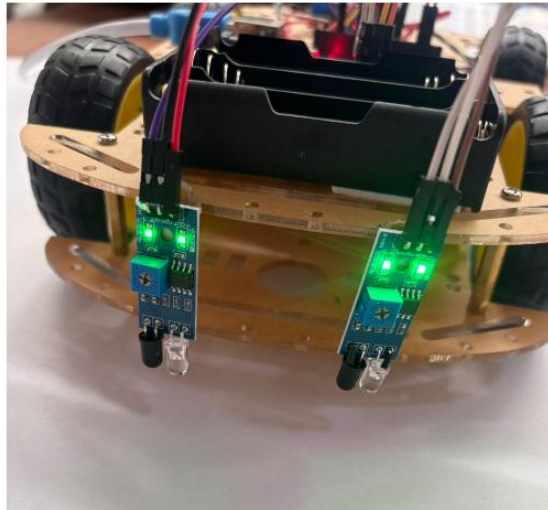


Figure 20: IR sensor, sensing white surface

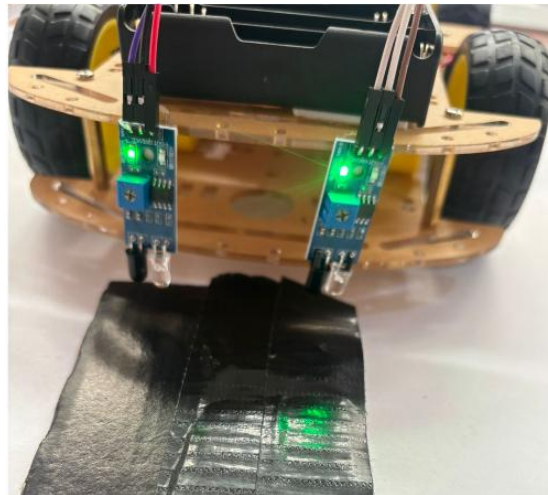


Figure 21: IR Sensor, sensing black surface

Step 12: Build the track

1. Get a large chart paper
2. Paste a black tape to make the path

3.2 Flowchart and Algorithm of the Code

3.2.1 Flowchart

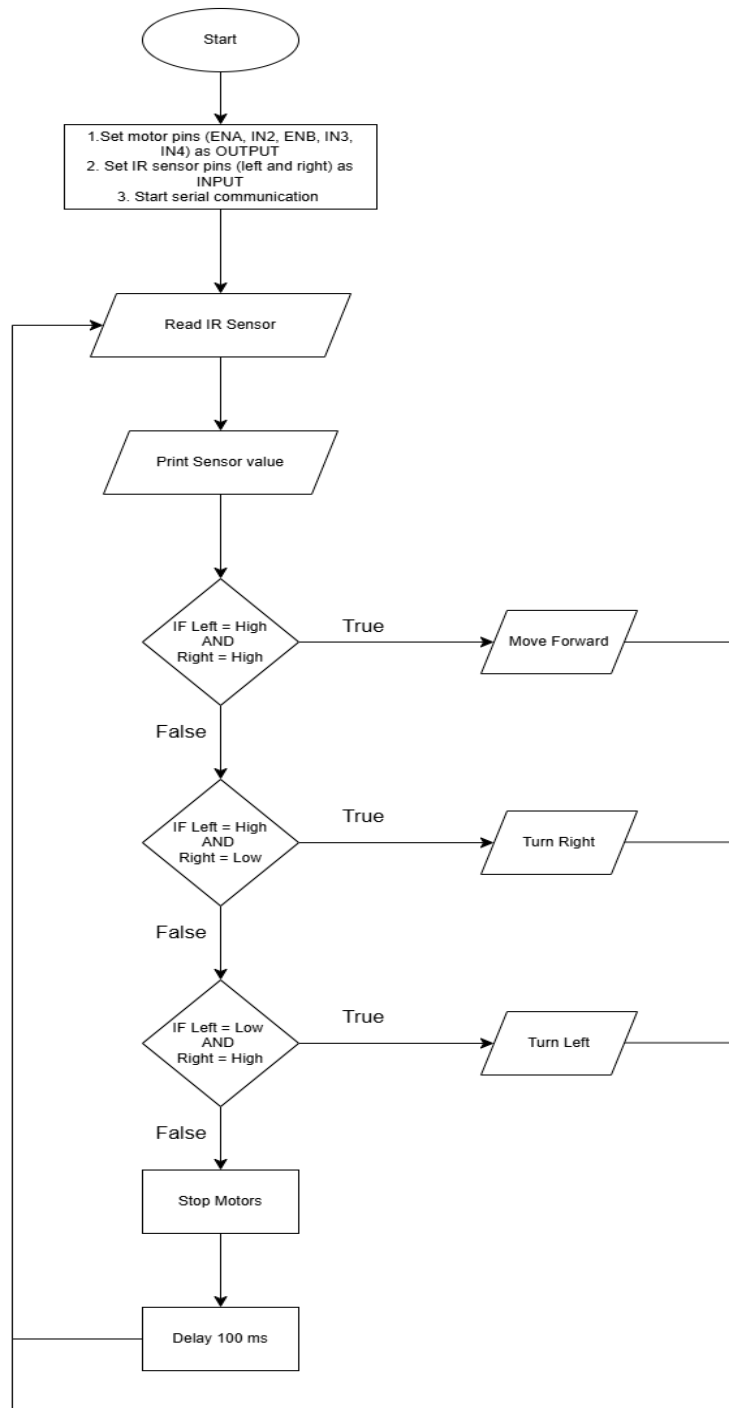


Figure 22 Flowchart of Code

3.2.2 Algorithm

Step1. Start

Step 2. Set motor pins (ENA, IN1, IN2, ENB, IN3, IN4) as OUTPUT

Step 3. Set IR sensor (left, right) as INPUT

Step 4. Start Serial communication

Step 5. Read IR sensor value and Print IR sensor value

Step 6. Check conditions:

- If Left = High AND Right = High, it moves Forward
- Else if Left = High AND Right = Low, it moves Right
- Else if Left = Low AND Right = High, it moves Left
- Else if both Low motors stop

Step 7. Goes to step 5 (loop continues infinity)

3.3 Software Process:

Step 1: Software Logic Design

Left Sensor	Right Sensor	Action
1 (Black)	1 (Black)	Forward
1 (Black)	0 (White)	Turn Right
0 (White)	1 (Black)	Turn Left
0 (White)	0 (White)	Stop

Table 2: Software Logic Design of IR Sensor

Step 2: Code Development

```
// ===== MOTOR A =====
```

```
const int ENA = 11;
```

```
const int IN1 = 10;
```

```
const int IN2 = 9;
```

```
// ===== MOTOR B =====
```

```
const int ENB = 6;
```

```
const int IN3 = 8;
```

```
const int IN4 = 7;
```

```
// ===== IR SENSORS =====
```

```
const int IRSensorLeft = 2;
```

```
const int IRSensorRight = 3;
```

```
// ===== SPEED =====
```

```
#define s 60
```

```
#define t 200
```

```
void setup() {
```

```
    // Motor pins
```

```
    pinMode(ENA, OUTPUT);
```

```
    pinMode(IN1, OUTPUT);
```

```
    pinMode(IN2, OUTPUT);
```

```
    pinMode(ENB, OUTPUT);
```

```
    pinMode(IN3, OUTPUT);
```

```
    pinMode(IN4, OUTPUT);
```

```
    // Sensor pins
```

```
    pinMode(IRSensorLeft, INPUT);
```

```
    pinMode(IRSensorRight, INPUT);
```

```
    // Serial monitor
```

```
    Serial.begin(9600);
```

```
}
```

```
void loop() {
```

```
L2N1
```

```
bool leftSensor = digitalRead(IRSensorLeft);

bool rightSensor = digitalRead(IRSensorRight);


// Debugging

Serial.print("Left: ");

Serial.print(leftSensor);


Serial.print(" | Right: ");

Serial.println(rightSensor);


// BOTH ON BLACK → FORWARD

if (leftSensor == HIGH && rightSensor == HIGH) {

    moveForward();

}


// RIGHT SENSOR OFF LINE → TURN RIGHT

else if (leftSensor == HIGH && rightSensor == LOW) {

    turnRight();

}


// LEFT SENSOR OFF LINE → TURN LEFT

else if (leftSensor == LOW && rightSensor == HIGH) {
```



```
    turnLeft();  
}  
  
// BOTH WHITE → STOP  
else {  
    stopMotors();  
}  
delay(10);  
}  
  
// ===== FORWARD =====  
void moveForward() {  
  
    // Motor A forward  
    digitalWrite(IN1, HIGH);  
    digitalWrite(IN2, LOW);  
    analogWrite(ENA, s);  
  
    // Motor B forward  
    digitalWrite(IN3, LOW);  
    digitalWrite(IN4, HIGH);  
    analogWrite(ENB, s);  
}
```

```
// ===== TURN RIGHT =====
```

```
void turnRight() {
```

```
    // Left motor forward
```

```
    digitalWrite(IN1, HIGH);
```

```
    digitalWrite(IN2, LOW);
```

```
    analogWrite(ENA, t);
```

```
    // Right motor stop
```

```
    digitalWrite(IN3, LOW);
```

```
    digitalWrite(IN4, LOW);
```

```
    analogWrite(ENB, 0);
```

```
}
```

```
// ===== TURN LEFT =====
```

```
void turnLeft() {
```

```
    // Left motor stop
```

```
    digitalWrite(IN1, LOW);
```

```
    digitalWrite(IN2, LOW);
```

```
    analogWrite(ENA, 0);
```

```
// Right motor forward

digitalWrite(IN3, LOW);

digitalWrite(IN4, HIGH);

analogWrite(ENB, t);

}


// ===== STOP =====

void stopMotors() {

    digitalWrite(IN1, LOW);

    digitalWrite(IN2, LOW);

    analogWrite(ENA, 0);

    digitalWrite(IN3, LOW);

    digitalWrite(IN4, LOW);

    analogWrite(ENB, 0);

}
```

4. Testing and Findings

4.1 Test 1: Straight line, Centered start

Objective:	To test if the robot moves straight without deviation when placed at the centre of a straight black line
Action:	1. Robot placed at the centre of a 1-meter straight black line on white surface 2. Power switched on
Expected Result:	Robot should move straight forward along the line without oscillating or leaving the line
Actual Result:	Robot moved smoothly straight along the line for the full 1 meter with no deviation
Conclusion	The test was successful.

Table 3: Straight line, centered start

4.2 Test 2: Straight line, Offset Left & Offset Right

Objective:	To see whether the robot self-corrects when offset from center on a straight line
Action:	Robot positioned with left side on the black line (offset left) - Robot positioned with right side on the black line (offset right) - Robot started in both positions
Expected Result:	If offset left, robot will correct by turning right. If offset right, robot will correct by turning left
Actual Result:	In both cases, the robot corrected itself in less than 0.2 seconds and continued on a straight line
Conclusion:	The test was successful.

Table 4: Straight line, Offset Left & Right

4.3 Test 3: Sharp 90° Left Curve

Objective:	To verify if the robot can perform a sharp 90° left turn without failing to detect the line.
Action:	• The robot was placed in front of a sharp 90° left turn (radius 25 cm) • Activated the robot and set the speed to normal (150 PWM) • Adjusted the speed to 120 PWM and observed
Expected Result:	The robot will be able to maneuver around the sharp bend and follow the line correctly.
Actual Result:	The robot did not manage to perform the sharp 90° left turn at both the given speeds. It failed to recognize the presence of the line during the maneuver.
Conclusion:	The experiment was a failure. The robot was not able to handle sharp turns due to its incapability of turning appropriately.

Table 5: Sharp 90° Left Curve

4.4 Test 4: T-Junction (End of Line)

Objective:	To see whether the robot halts upon reaching a T-junction or end of track
Action:	- The robot is positioned on the track at the point where a T-junction exists - The robot is powered up and allowed to proceed
Expected Result:	When both IR sensors detect the black line simultaneously, the robot should stop immediately
Actual Result:	The robot will automatically halt once both the IR sensors pick the black strip at the same time
Conclusion:	The test was successful.

Table 6: T-Junction (End of Line)

4.5 Test 5: IoT Sensor Output Monitoring

Objective:	Test if the robot can provide output from sensors to integrate into IoT Cloud
Action:	- The robot was connected to the computer using USB cable - Opened Arduino Serial Monitor - Powered on the robot and made it follow the black line
Expected Result:	Expected Left and Right sensors data will print in CSV form like ("0,1", "1,0") each loop
Actual Result:	The sensor output was printing successfully in the Serial Monitor while the robot was following the black line
Conclusion:	Test completed successfully. Output can be used for future cloud integration through ESP32.

Table 7: IoT Sensor Output Monitoring

5. Further work

Even though the current prototype can successfully track the black line, there are certain areas that could be improved to increase the robot's efficiency and intelligence. The first thing is to implement a PID controller (proportional-integral-derivative) which will provide the robot smooth transition while turning and driving along the black line instead of relying in if-else condition. This will make the robot more stable, accurate and faster. The PID controller can also make calculations on how far the line is and the robot can accordingly make appropriate corrections and movements. The second upgrade would be to add ESP module to the vehicle for Wi-Fi connections and cloud connection. Connecting the robot with cloud would allow remote control over some features of the robot.

6. Conclusion

Conclusively, the current project has explored the design of an autonomous car that can follow lines with high accuracy. The project has emphasized the significance of such an autonomous car in enhancing the efficiency of movement in today's automation field where the use of autonomous systems is increasingly on the rise.

A comprehensive problem statement and objective setting were carried out to identify the problem area and its solution. A system design was then proposed which included various diagrams to highlight the interplay between the hardware and software parts. The system requirement analysis revealed the key hardware and software requirements that would be needed to develop the system successfully.

The project has followed an Agile development process along with using prototypes in its implementation, and testing of the system at hand.

Finally, the current project has demonstrated an application of embedded systems and robotics concepts to develop an intelligent autonomous car. Moreover, the project will provide experience about various engineering-related issues while developing technical and team.

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